Lab 2 – Design a PoW Consensus

Information Security UESTC4036, School of Engineering, University of Glasgow

Student Name: _____

Student Number: _____

UESTC4036 Lab Assessment

Each individual lab will be evaluated using the below criteria:

- Design a PoW consensus in Python– 50%

  This means each student will need to demonstrate a functional blockchain system for each task in the lab manual to the GTA and member of staff to get the full mark.
- Lab Questions – 10%
- Reflection and significance – 25%

  After the completion of programming and testing each of the “Tasks”, you will be required to reflect, in a few lines, on your understanding of the tasks and its significance. This is a chance for you to ask yourselves “What you did?”, “What it means?”, and “What next?”, meaning what is the significance of the undertaken task in the context of blockchain and what you are learning in the lecture. Try to be innovative and think outside the box. Please, do not provide a summary of what you did as it will not be accepted.
- Adherence to coding good practice – 15%
  - Meaningful variable and function names – please avoid names like “x”, “y”, or personal names
  - Using annotate comments to explain what your code does at the beginning and to clarify what each important line of code does.
  - Consistent indentation – Indentation helps in making the code readable and to avoid error.

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1. Lab 2 Objective

The objective of this lab is to design and test blockchain technology and understand its components in detail. A Proof of Work (PoW) consensus is used in this blockchain. In

this lab, we will continue the blockchain design of Lab 1.

2. Software Requirements

Software:

- **Python**
- **Visual Studio Code**

3. Preliminary

Login to your account on the website https://besharing.net/. Check other students' ideas about the PoW consensus design.

4. Task 3: Proof of Work

Write a proof of work function. In the *generate_blockchain()* function, we had a variable called proof. This variable represents the proof of work done to mine a block.
Task3.1: As the programmer of the blockchain, please create an algorithm that the miners will solve to mine a block successfully. Please start by creating a new method called *proof_of_work().*

Task3.2: Use JSON dumps to encode the block and return a cryptographic hash, *hashlib.sha256()* is used to calculate the hash value.

Task3.3: In the method, please create a variable to store the proof submitted by miners, and create a control statement to check the status of the proof of work, which by default will be *False*.

Task3.4: Create a new variable called *hash_operation*, which is used to encode the problem in a cryptographic hexadecimal digit with the use of the SHA256 hash library. Note that hash value does not need to be fully verified, i.e.
if hash_operation[:4] == "0000":

5. Mine New Block

Active the virtual environment, import the *Blockchain* class and use the mine block function to add two blocks to the blockchain.

6. Validate the Blockchain (Optional)

In this step, please create a function that checks if the entire blockchain is valid. This step is crucial in maintaining the integrity of our blockchain to ensure that none of our blocks is corrupt. Check if the previous hash of the current block is the same as the hash

of it's previous block.

7. Task 4: Reflection

After the completion of programming and testing please reflect, in a few lines, on your understanding of the tasks and its significance. This is a chance for you to ask yourselves “What you did?”, “What it means?”, and “What next?”, meaning what is the significance of the undertaken task in the context of PoW consensus and what you are learning in the lecture. Try to be innovative and think outside the box. Please, do not provide a summary of what you did as it will not be accepted. **[Answer in the below space]**

8. Lab Score

Marking Criteria	**Student’s Score**
Demonstration (50%)	
Lab Questions (10%)	
Reflection (25%)	
Adherence to coding good practice (15%)	